

LOKESH VARMA KONDURU

(+1) 571-244-4062 | lokeshvarmak@vt.edu | [linkedin.com/in/lokeshvarmakonduru](https://www.linkedin.com/in/lokeshvarmakonduru) | github.com/LokeshVarma-Konduru

SUMMARY

Software Engineer with 3+ years of professional experience architecting full-stack platforms, cloud-native **microservices**, and production-grade **GenAI** systems, currently pursuing an M.S. in Computer Science at Virginia Tech (GPA: 4.0/4.0). Proficient in **React**, **Node.js**, **Python**, **Java/Spring Boot**, and **AWS** infrastructure (**EC2**, **SageMaker**, **Kubernetes**, **Kafka**). Skilled in designing **LLM-powered RAG** pipelines, AI agent workflows, and scalable distributed systems across agri-tech, fraud detection, and e-commerce domains.

EDUCATION

Virginia Tech | Master's in Computer Science | GPA: 4.0/4.0 | Virginia, USA

May 2026

WORK EXPERIENCE

Software Engineer Virginia Tech Sept 2024 – May 2026

- Architected a full-stack platform using **React/TypeScript/JavaScript**, **Next.js SSR**, **Redux Toolkit**, **React Query**, **Tailwind CSS**, **Node.js/Express REST APIs**, and **PostgreSQL** relational data layer, with **Jest** unit and integration tests cutting bug turnaround by **40%**.
- Containerized **React/TypeScript** frontend and **Node.js/Express** backend with **Docker** and **Kubernetes**, deployed to **AWS EC2**, configured **Amazon S3** for production data and logs, and automated **CI/CD** pipelines via **GitHub Actions** with **Terraform** IaC.
- Engineered a production-grade **GenAI** advisory assistant integrating **GPT-4o**, **LangChain**, **LangGraph**, **FAISS** vector database, and **OpenAI** embeddings into **Node.js/Express REST API** services, designing **LLM-powered AI agent pipelines** with **Retrieval-Augmented Generation (RAG)**, **semantic chunking**, **hybrid vector retrieval**, and **MCP tool calling** for **real-time** AI-powered recommendations.
- Built 5+ **React/TypeScript/JavaScript**, **HTML5/CSS3** analytics tools using **Google Earth Engine APIs**, consolidated into a **React/Tailwind CSS** platform with **Firebase** telemetry and a **Prometheus/Grafana** observability dashboard, improving issue resolution by **20%**. [website](#), [VT Publications Profile](#)
- Developed a **Python/FastAPI/Panel** geospatial application integrated with **Google Earth Engine REST APIs**, building **FastAPI** backend services with **Redis** caching, **asyncio-based async processing**, and **Python data validation** pipelines, improving processing efficiency by **40%**.

Software Engineer (Contract) Marine Corps Community Services (MCCS) Sept 2024 – Dec 2025

- Architected an AI-powered fraud detection platform with **React/TypeScript/JavaScript** frontend, **Tailwind CSS**, **Node.js/Express REST APIs**, **MongoDB** data layer, and **Redis** caching for fraud score and transaction lookups across **microservices** handling **10K+** transactions.
- Built a **GenAI** fraud investigation assistant integrating **Gemini**, **LangChain**, **LangGraph**, **FAISS** vector database, and **OpenAI** embeddings into **Node.js/Express REST API** services, designing **LLM-powered AI agent pipelines** with **Retrieval-Augmented Generation (RAG)**, **semantic chunking**, **prompt engineering**, **MCP tool calling**, and **hybrid vector retrieval**, cutting manual fraud review time by **30%**.
- Orchestrated **microservices** with **Kubernetes** and **Docker** on **AWS EC2**, streaming **real-time** transactions via **Kafka** for **event-driven** fraud signal processing, integrating **Amazon SageMaker** for ML model training and batch inference, **CloudWatch** observability, and **OpenTelemetry** distributed tracing, storing model artifacts and inference logs in **Amazon S3**. [Github](#)

Software Engineer DuoDecimal Tech LLP July 2023 – July 2024

- Engineered full-stack platforms for events, bookings, hostel, and food management using **React/TypeScript/JavaScript**, **HTML5/CSS3**, **Node.js/Express REST APIs**, **MongoDB** data layer, and **JWT/OAuth2** authentication across **React** frontend components and **microservices** data layers.
- Built and scaled a **React/TypeScript/JavaScript** conference booking and food ordering platform with **Razorpay** and **Easebuzz** payment integrations, **Socket.io WebSocket** real-time ordering, **Node.js/Express REST API** backend services, **Redis** for distributed session and cart caching, and **Kafka** for **event-driven** async order processing, improving booking efficiency by **30%** and reducing response latency by **50%**.
- Developed a hostel and guest management system integrated with **RFID** devices, building **Node.js/Express REST APIs** and **MongoDB** data layer to automate **real-time** access control and **event-driven** workflows, reducing manual paperwork by **60%** for **25,000+** users.
- Deployed full-stack **React/TypeScript/JavaScript** **microservices** on **AWS EC2** behind **AWS ALB** load balancer, routing **Node.js/Express REST APIs** through **AWS API Gateway**, and maintaining **CI/CD** pipelines via **GitHub Actions** with **Jest** automated testing for **25,000+** users.

Software Engineer Intern MindGraph Technologies Jan 2023 – June 2023

- Built client-facing dashboards and **Python/Flask** backend REST API services using **Plotly Dash**, **pandas**, **NumPy**, and **Python** data visualization pipelines for Telecom and CPG clients, integrating **Azure Active Directory** SSO and **OAuth2**, containerizing with **Docker**, storing client data in **Azure Blob Storage**, and deploying on **Microsoft Azure** with **Azure DevOps CI/CD**, improving data visibility by **40%**.
- Designed and optimized **distributed batch and streaming data pipelines** processing **10M+** airline records using **SQL**, **Spark SQL**, **PySpark** UDFs, and **Kafka** for **event-driven real-time** streaming, orchestrating **Python** ETL workflows with **Apache Airflow** and automated scheduling, and implementing automated unit and integration testing with **PyTest** to ensure data quality, data observability, performance, and reliability.

PROJECTS

Bookstore AI E-Commerce Platform | React, Java/Spring Boot, MySQL Feb 2025 – May 2025

- Built an AI-powered bookstore platform using **React/TypeScript/JavaScript**, **Java/Spring Boot**, **Spring Security**, **MySQL**, and **Redis** caching, implementing **JWT/OAuth2** authentication, **microservices**-ready REST APIs, catalog and cart workflows, and scalable user flows.
- Integrated a **GenAI** RAG recommendation assistant using **Claude**, **LangChain**, **LangGraph**, **FAISS** vector database, and **OpenAI** embeddings with **semantic chunking** and hybrid retrieval to support semantic search, personalized recommendations, and product discovery.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, C, C++, HTML5, CSS

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Redux Toolkit, React Query, Tailwind CSS, Spring Boot, Spring Security, Flask, FastAPI, Socket.io, Plotly Dash, PySpark

AI & GenAI: LLMs, RAG, LangChain, LangGraph, FAISS, Pinecone, OpenAI Embeddings, Semantic Chunking, Vector Search, Prompt Engineering, AI Agents, MCP, MLOps, GPT-4o, Claude, Gemini

Cloud & DevOps: AWS EC2, Amazon S3, Amazon SageMaker, AWS ALB, AWS API Gateway, AWS CloudWatch, Microsoft Azure, Terraform, Docker, Kubernetes, Kafka, Apache Airflow, CI/CD Pipelines, GitHub Actions, OpenTelemetry

Databases & Caching: MongoDB, PostgreSQL, MySQL, Redis, Firebase

Tools & Testing: Git, GitHub, REST APIs, JWT, OAuth2, Azure Active Directory, Google Earth Engine, Jest, PyTest, TDD, Agile/Scrum, Katalon Studio, Jira, Microservices, Distributed Systems, Spark SQL